

Mahir Puri

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EDUCATION

University of Victoria

Victoria, BC

Bachelor of Software Engineering (B.Seng) – GPA: 4.0 / 4.0

Sept. 2024 – Dec. 2028

- **Relevant coursework:** Data Structures & Algorithms, Distributed Systems, Database Systems, Object-Oriented Design, Software Testing, Statistics
- **Awards:** RBC Global Security Student Innovation Challenge (2026) | Certificate of Achievement, UVic Mathematics & Statistics (2025) | Scholarship for Academic Excellence (2024)

EXPERIENCE

Returning Software Engineering Co-op

Sept. 2026 – Dec. 2026

Royal Bank of Canada (RBC) – Commercial Core Banking & Payments Technology

Toronto, ON

- Selected to return to **RBC's Real-Time Payments Cloud Team** to support Java and Spring Boot payment APIs on OpenShift within production-critical **financial infrastructure**.
- Expected to build reusable service components for real-time integrations across payment systems, with automated testing, pair programming, and peer review supporting reliable delivery.

AI & Software Engineer Co-op

Jan. 2026 – Apr. 2026

Royal Bank of Canada (RBC) – Global Security, Cyber Technology Office

Toronto, ON

- Architected **Medallion ETL pipelines** on Azure Databricks with PySpark and Delta Lake -integrated **4 enterprise security sources** across **500+ applications** with incremental loading, schema validation & data-quality gates.
- Diagnosed async ingestion and data-integrity failures by tracing Databricks job logs and Delta transaction history – root-caused an **SCD Type 2 phantom-update defect** causing incorrect downstream state.
- Built **FastAPI + GraphQL** service layers exposing normalized security data across pipeline stages; designed **SQL models** tracking vulnerability counts, SLA breaches, and risk scores.
- Contributed to **69% faster remediation** across the application portfolio and eliminated **96+ hours of monthly overhead** by automating reporting workflows in Python with idempotent, re-runnable pipeline design.
- Served as primary technical contact for **3–4 internal application teams** – coordinated data delivery, participated in peer code reviews, and documented system behavior in Confluence.
- Won the **RBC Global Security Student Innovation Challenge** – presented platform architecture to Directors, Senior Directors, and the **Chief Security Officer**.

PROJECTS

RTPN – Real-Time Payments Network Simulator | *Java, Spring Boot, Kafka, PostgreSQL, Docker* | [GitHub](#)

2026

- Designed a **double-entry RTGS ledger** in PostgreSQL with idempotency-keyed debit and credit entries to preserve balance invariants under concurrent payment requests.
- Implemented a Kafka message rail with consumer retries and **dead-letter handling**; validated reconciliation behavior using a Python simulator generating concurrent multi-institution traffic – **zero inconsistencies** in test runs.
- Extended with a **read-only AI operations agent** using Anthropic tool-use that lets engineers query payment state in plain language without touching the settlement write path.

Anchor – Durable Agent Runtime | *Python, FastAPI, PostgreSQL, DynamoDB, SQS* | [GitHub](#)

2026

- Designed event-sourced replay and idempotent side-effect handling so interrupted workflows resume without duplicating completed operations – validated through **69 automated tests** covering process termination, replay, and state recovery.
- Implemented dual persistence backends – PostgreSQL for local development and DynamoDB + SQS for cloud deployment – with a FastAPI control plane exposing workflow status, replay triggers, and side-effect logs.

Ballast – Kubernetes Inference Autoscaler | *Go, Kubernetes, controller-runtime, Helm* | [GitHub](#)

Ongoing

- Building a Go Kubernetes operator with a custom InferenceService CRD that autoscales LLM serving deployments on **pending + in-flight requests per replica** instead of CPU – with scale-down stabilization and rolling model updates.
- Shipping full Helm chart, RBAC manifests, envtest integration tests, and GitHub Actions CI – Docker Compose stack with a FastAPI mock server for GPU-free local testing.

TECHNICAL SKILLS

Languages: Python, Java, Go, SQL, Bash

Backend, APIs & Storage: FastAPI, Spring Boot, GraphQL, REST APIs, PostgreSQL, DynamoDB, MongoDB, MySQL

Data & Streaming: Kafka, PySpark, Azure Databricks, Delta Lake, ETL/ELT pipelines, SCD Type 2, schema validation

Cloud & Infrastructure: Kubernetes, controller-runtime, Helm, RBAC, Docker, AWS, OpenShift, Terraform, GitHub Actions

Testing & Reliability: Pytest, JUnit, Testcontainers, envtest, integration testing, event-sourced replay, idempotent processing, root-cause analysis

AI-Assisted Development: Claude Code, Anthropic tool-use agents